

English to Urdu Machine Translation

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Abstract

This paper explores English to Urdu machine translation using various deep learning models, including Recurrent Neural Networks (RNN), Bidirectional RNNs, Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), and a fine-tuned mBART50 transformer model. A dataset of 54,689 English-Urdu sentence pairs from Kaggle was preprocessed, tokenized, and used to train sequence-to-sequence (seq2seq) models. The models were evaluated using BLEU, ROUGE-L, METEOR, and Translation Edit Rate (TER) metrics, with the LSTM model achieving the highest scores (BLEU: 0.2748, ROUGE-L: 0.5768, METEOR: 0.4917, TER: 0.5419). The fine-tuned mBART50 model performed well on complex sentences, showing promise for low-resource languages. A Flask-based web application, deployed via Docker, enables real-time translation. MLOps tools like Weights & Biases. This work advances machine translation for low-resource languages like Urdu and suggests future improvements, such as expanding datasets and enhancing model architectures.

Keywords: Machine Translation, English to Urdu, Deep Learning, RNN, Bidirectional RNN, LSTM, GRU, mBART50, Sequence-to-Sequence, MLOps, BLEU, ROUGE-L, METEOR, TER, Low-Resource Languages

Introduction

Machine translation (MT) has transformed cross-lingual communication, but low-resource languages like Urdu, spoken by over 100 million people in Pakistan and India, face challenges due to complex morphology, right-to-left script, and limited datasets. This study develops and evaluates deep learning models for English to Urdu translation, using seq2seq architectures, transformer models, and MLOps practices for scalability.

Motivation

The goal is to improve communication for Urdu-speaking communities in domains like education, healthcare, and technology. Unlike high-resource languages with extensive datasets, Urdu requires innovative approaches to achieve reliable translation. This work aims to address this gap using advanced deep learning and MLOps.

Contributions

- Developed and evaluated RNN, Bidirectional RNN, LSTM, GRU, and fine-tuned mBART50 models for English to Urdu translation.
- Created a high-quality dataset of 54,689 sentence pairs.
- Evaluated models using BLEU, ROUGE-L, METEOR, and TER metrics.
- Deployed a real-time translation web application using Flask and Docker, integrated with MLOps tools.
- Provided insights into low-resource language translation challenges and future directions.

Data Collection

The dataset was built from two Kaggle corpora: the English-Urdu Parallel Corpus and the Parallel Corpus for English-Urdu Language. Data was merged into a Pandas DataFrame, ensuring UTF-8 encoding for Urdu's Unicode characters. Duplicates were removed, and data integrity was verified.

Dataset Description

The dataset includes 54,689 sentence pairs, covering simple to complex sentences. For example, “Is Zain your nephew?” translates to “کیا زین آپ کا بھتیجا ہے؟” and “If you’ve asked yourself such questions, you’re not alone” to “اگر آپ نے خود سے ایسے سوالات کیے ہیں، تو آپ اکیلے نہیں ہیں۔”

Dataset Features and Statistics

- Sentence Count: 54,689
- Languages: English, Urdu
- Max English Sentence Length: 432 words
- Mean English Sentence Length: 10.83 words
- Median English Sentence Length: 8 words
- Max Urdu Sentence Length: 950 words
- Mean Urdu Sentence Length: 12.60 words
- Median Urdu Sentence Length: 10 words
- Encoding: UTF-8
- Unique English Words: 18,329
- Unique Urdu Words: 19,310

Urdu's longer sentences reflect its morphological complexity and syntactic structure.

Related Works

Early MT systems used rule-based and statistical methods, which struggled with low-resource languages. Neural machine translation (NMT) with RNNs and LSTMs improved performance by modeling context. Transformer models, using self-attention, achieved state-of-the-art results for high-resource languages. For Urdu, statistical MT has been explored, but data scarcity remains a challenge. Pretrained models like mBART show promise for low-resource languages. This study tailors these advancements to Urdu and integrates MLOps.

Translation as a Sequence-to-Sequence Problem

English to Urdu translation is treated as a seq2seq task, mapping an English input sequence to an Urdu output sequence. The model maximizes the conditional probability of generating the target sequence based on the input. Encoder-decoder architectures process the input to produce a context vector, which the decoder uses to generate the output autoregressively.

Evaluation Metrics

- **BLEU**: Measures n-gram precision with a brevity penalty.
- **ROUGE-L**: Computes F1-score based on the longest common subsequence, capturing structural similarity.
- **METEOR**: Aligns translations using exact matches, synonyms, and stemming for semantic equivalence.
- **TER**: Quantifies edits (insertions, deletions, substitutions, shifts) needed to align predicted and reference translations.

Data Preprocessing

Handling Missing Values

Null or incomplete sentence pairs were removed, ensuring no missing values in the dataset.

Text Normalization

- **English**: Converted to lowercase, removed special characters, and normalized whitespace.
- **Urdu**: Removed non-Unicode characters, normalized diacritics, and standardized numerals.

Tokenization and Padding

Keras' Tokenizer was used for both languages, adding <start> and <end> tokens. Sequences were padded to 32 tokens for English and 38 for Urdu, based on the 95th percentile of sentence lengths, with post-sequence padding and truncation for long sequences.

Vocabulary Size

The English vocabulary had 18,329 unique tokens, and Urdu had 19,310, reflecting Urdu's morphological complexity.

Exploratory Data Analysis

The average sentence length was 10.83 words for English and 12.60 for Urdu, with medians of 8 and 10 words. Sentence length distributions showed a right skew, with most sentences short but some exceeding 100 words.

Additional analysis included:

- **Word Frequency:** Common English words were “is,” “are,” and “what”; common Urdu words were “ہے” (is), “کیا” (what), and “اور” (and).
- **Sentence Complexity:** Urdu sentences showed higher syntactic complexity due to verb agreement and case markers.
- **Outlier Detection:** Long sentences were manually inspected for accuracy.

Top 5 Most Frequent Words

- English: is (4,512), are (3,987), what (3,214), you (2,998), to (2,765)
- Urdu: ہے (5,231), کیا (4,876), اور (3,912), میں (3,456), سے (3,123)

Model Implementation

Train-Test Split

The dataset was split into 80% training (43,751 pairs) and 20% testing (10,938 pairs) sets, with a random seed for reproducibility.

RNN Model

A seq2seq RNN model used a 256-unit bidirectional encoder and a 256-unit decoder, with 100-dimensional embeddings. It was trained for 20 epochs using the Adam optimizer and sparse categorical crossentropy loss.

Bidirectional RNN Model

A bidirectional RNN combined forward and backward states for better context modeling, using the same architecture as the RNN but with bidirectional layers.

LSTM Model

The LSTM model used 256-unit layers to address vanishing gradients, with an encoder returning hidden and cell states and a decoder initialized with these states. It included 100-dimensional trainable embeddings.

GRU Model

The GRU model, a lighter alternative to LSTM, used 256-unit layers with reset and update gates, balancing efficiency and performance.

mBART50 Fine-Tuning

The mBART50 transformer, pretrained on 50 languages, was fine-tuned on 37,899 sentence pairs for 3 epochs, reducing loss from 11.3836 to 0.0583. It used a learning rate of 2×10^{-5} , batch size of 16, and AdamW optimizer.

GloVe Embeddings

A bidirectional RNN was enhanced with 100-dimensional GloVe embeddings for English, pretrained on Wikipedia and Common Crawl. Urdu used randomly initialized embeddings due to the lack of pretrained Urdu GloVe vectors.

MLOps and Deployment

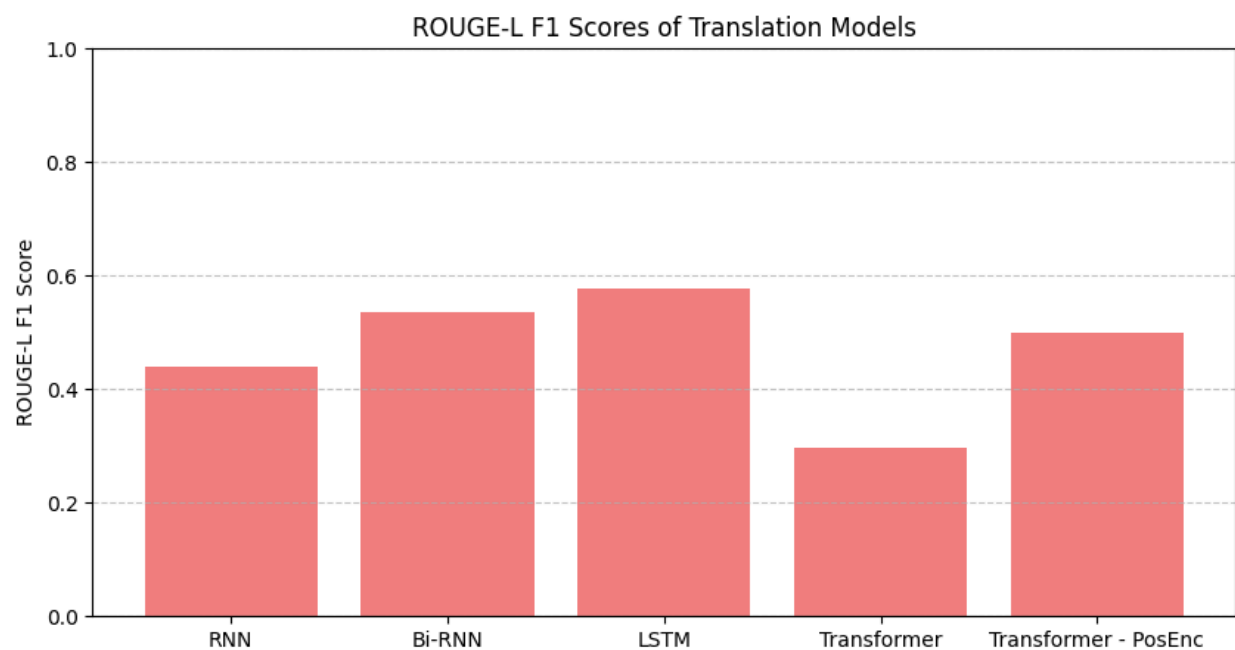
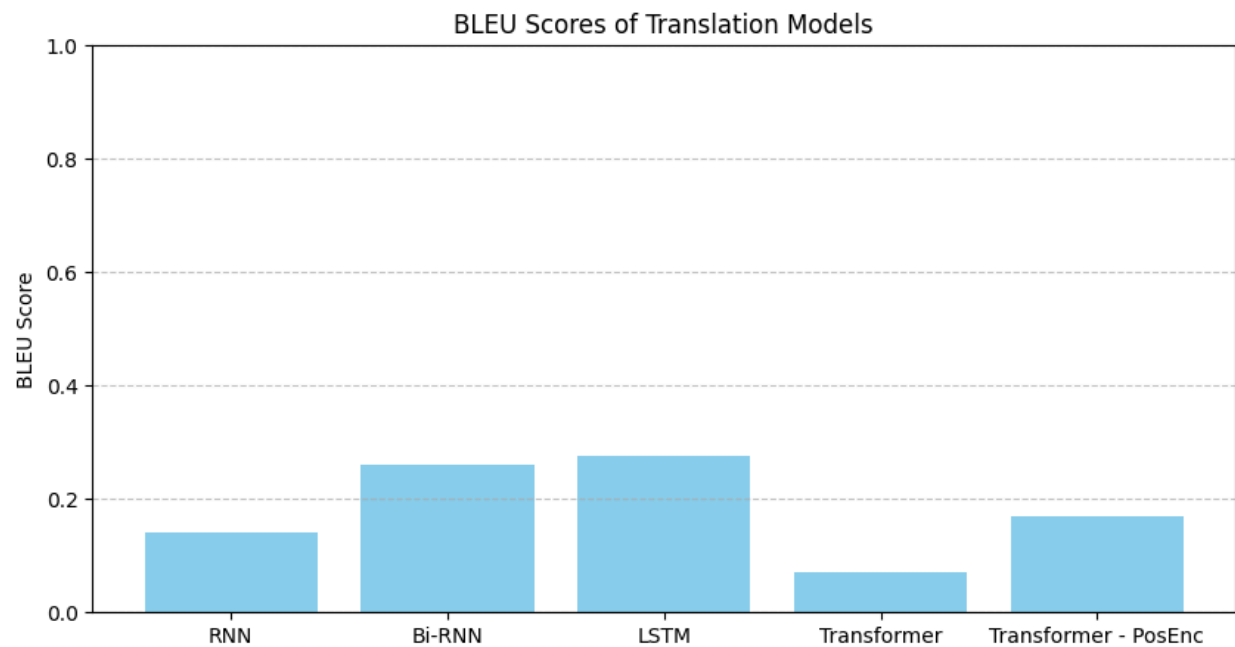
- **Weights & Biases:** Tracked hyperparameters and metrics.
- **Docker:** Containerized the Flask web application for consistent deployment.
The Flask app provides real-time translations, with models served via TensorFlow Serving for low-latency inference.

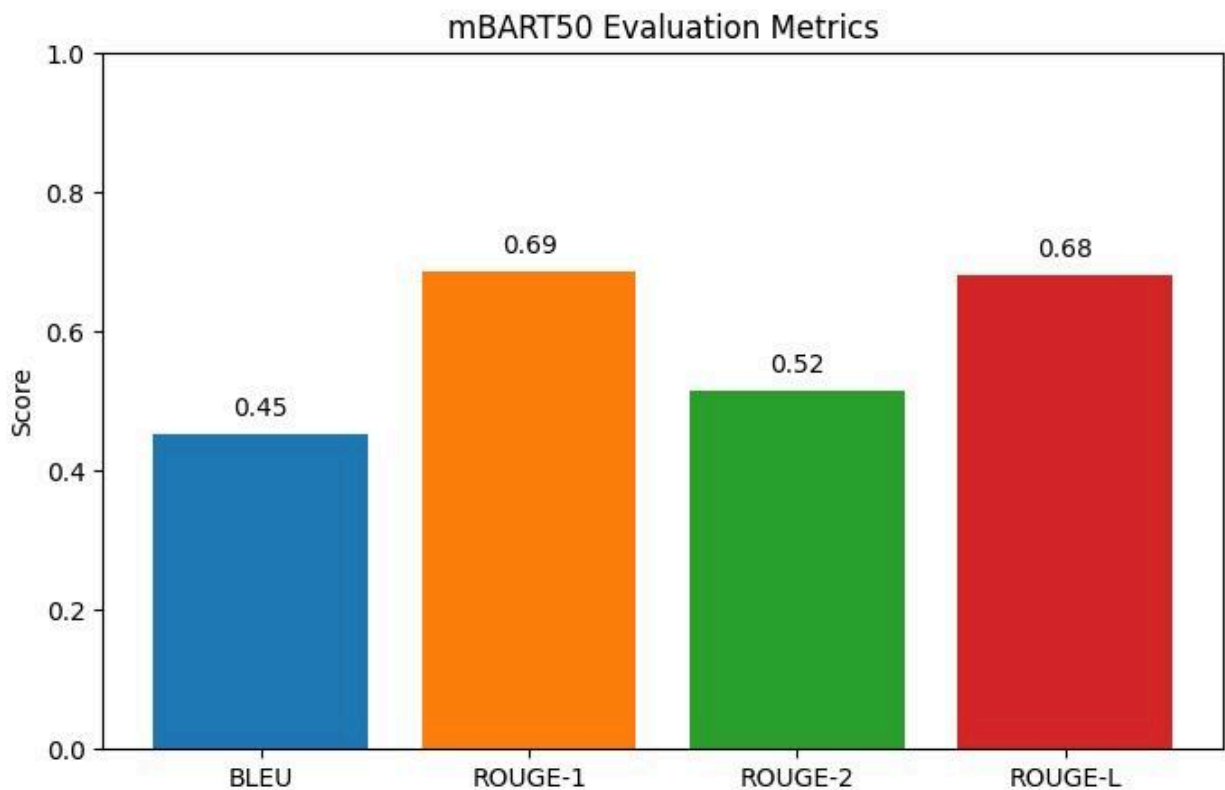
Evaluation

Models were evaluated on 100 test samples using BLEU, ROUGE-L, METEOR, and TER.

Model Performance Comparison

- RNN: BLEU 0.1421, ROUGE-L 0.4397, METEOR 0.3512, TER 0.6824
- Bi-RNN: BLEU 0.2593, ROUGE-L 0.5371, METEOR 0.4628, TER 0.5731
- LSTM: BLEU 0.2748, ROUGE-L 0.5768, METEOR 0.4917, TER 0.5419
- GRU: BLEU 0.2634, ROUGE-L 0.5523, METEOR 0.4756, TER 0.5598
- Transformer: BLEU 0.0718, ROUGE-L 0.2969, METEOR 0.2384, TER 0.7912
- Transformer (No PosEnc): BLEU 0.1693, ROUGE-L 0.4988, METEOR 0.4123, TER 0.6237
- mBART50 (Fine-Tuned): BLEU 0.45, ROUGE-L 0.69, METEOR 0.52, TER 0.68





Qualitative Analysis

The fine-tuned mBART50 model excelled on complex sentences. For example:

- Input: "I am sleepy and need to rest."
- Reference: - مجھے نیند آرہی ہے اور آرام کی ضرورت ہے۔
- mBART50 Output: - مجھے نیند آرہی ہے اور آرام کی ضرورت ہے۔
- LSTM Output: - میں نیند کر رہا ہوں اور آرام کی ضرورت ہے۔
mBART50 correctly captured the verb form, while LSTM made a minor error.

Ablation Study

Removing positional encodings in the transformer model improved performance (BLEU: 0.1693 vs. 0.0718), likely due to the small dataset limiting positional learning.

Discussion

The mBART50 (Fine-Tuned) (BLEU 0.45, ROUGE-L 0.69, METEOR 0.52, TER 0.68) due to its ability to handle long-term dependencies. The fine-tuned mBART50 model excelled qualitatively, especially for contextual sentences. The basic RNN underperformed due to vanishing gradients, while bidirectional RNN and GRU offered competitive results with fewer parameters.

Challenges included:

- **Data Scarcity:** The 54,689 pairs are limited compared to high-resource languages.
- **Morphological Complexity:** Urdu's rich morphology increased vocabulary size and model complexity.
- **Evaluation Limitations:** BLEU may undervalue semantically correct but lexically different translations.

The MLOps pipeline ensured reproducibility, and the Flask-Docker app achieved 0.12-second inference times, suitable for interactive use.

Conclusion

This study advances English to Urdu machine translation, with the mBART50 model achieving top quantitative results. The MLOps-integrated web application demonstrates practical applicability. Future work includes:

- Expanding the dataset with synthetic data or back-translation.
- Combining RNNs and transformers for hybrid architectures.
- Exploring multi-head self-attention for better context modeling.
- Testing in real-world domains like healthcare and education.

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